Aim: Simulate supervised and unsupervised learning with simple python examples.

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F - 02

SUPERVISED LEARNING

In [3]:
```
import numpy as np
from sklearn.linear_model import LinearRegression


X = np.array([
    [400, 2],
    [800, 3],
    [1000, 3],
    [1200, 4],
    [1800, 5]
])

y = np.array([100000, 160000, 200000, 240000, 300000])

model = LinearRegression()
model.fit(X,y)

new_house = [[1100, 4]]
predicted_price = model.predict(new_house)

print("Predicted house price:", int(predicted_price[0]))
```

```
Predicted house price: 215000
```

In [8]:
```
import pandas as pd

data = pd.read_csv("ML Exp 2.csv")

print(data.head())
print(data.columns)
```

```
   size  rooms   price
0    50      2  150000
1    70      3  200000
2    80      3  250000
3    60      2  180000
4    50      4  300000
Index(['size', 'rooms', 'price'], dtype='object')
```

In [12]:

```python
import pandas as pd
from sklearn.linear_model import LinearRegression

data = pd.read_csv("ML Exp 2.csv")
print(data.columns)

X= data[["size","rooms"]]
y= data["price"]

model = LinearRegression()
model.fit(X,y)

new_house = pd.DataFrame({
    "size" : [1100],
    "rooms" : [4]
})

predicted_price = model.predict(new_house)

print("Predicted house price:", int(predicted_price[0]))
```

```
Index(['size', 'rooms', 'price'], dtype='object')
Predicted house price: 612315
```